

Introduction – A Virtual Pond

- **The Environment:** A simulated aquaculture pond for Largemouth Bass.
- **Dynamics:** Cyclic interactions between fish, food, probiotics, oxygen, NH3, disease, and parasites.
- **Goal:** Replacing original Evolutionary Algorithm with LLM to optimize the policies (of food, probiotic, and oxygen) to reach the highest Fitness in 30 simulation days.

OPTIMIZATION METRIC (THE GOAL)

$$\text{Fitness} = 0.55 \times \text{Yield} + 0.33 \times \text{SavingRate} + 0.12 \times \text{Healthiness}$$

The LLM agent must balance these three competing objectives.

LIVE SIMULATION: <https://jam.dev/c/6d18c995-6f4b-4d9a-9e1d-28bdda3d040b>

Technical Approach – Modular Architecture

We use 4 toggleable cognitive modules to test different configurations to measure the impact of each module.



Observation (Obs)

Reads pond state (water quality, health, yield) in natural language.



Tool (Tool)

Executes actions and calculates critical budget constraints.



Planning (Plan)

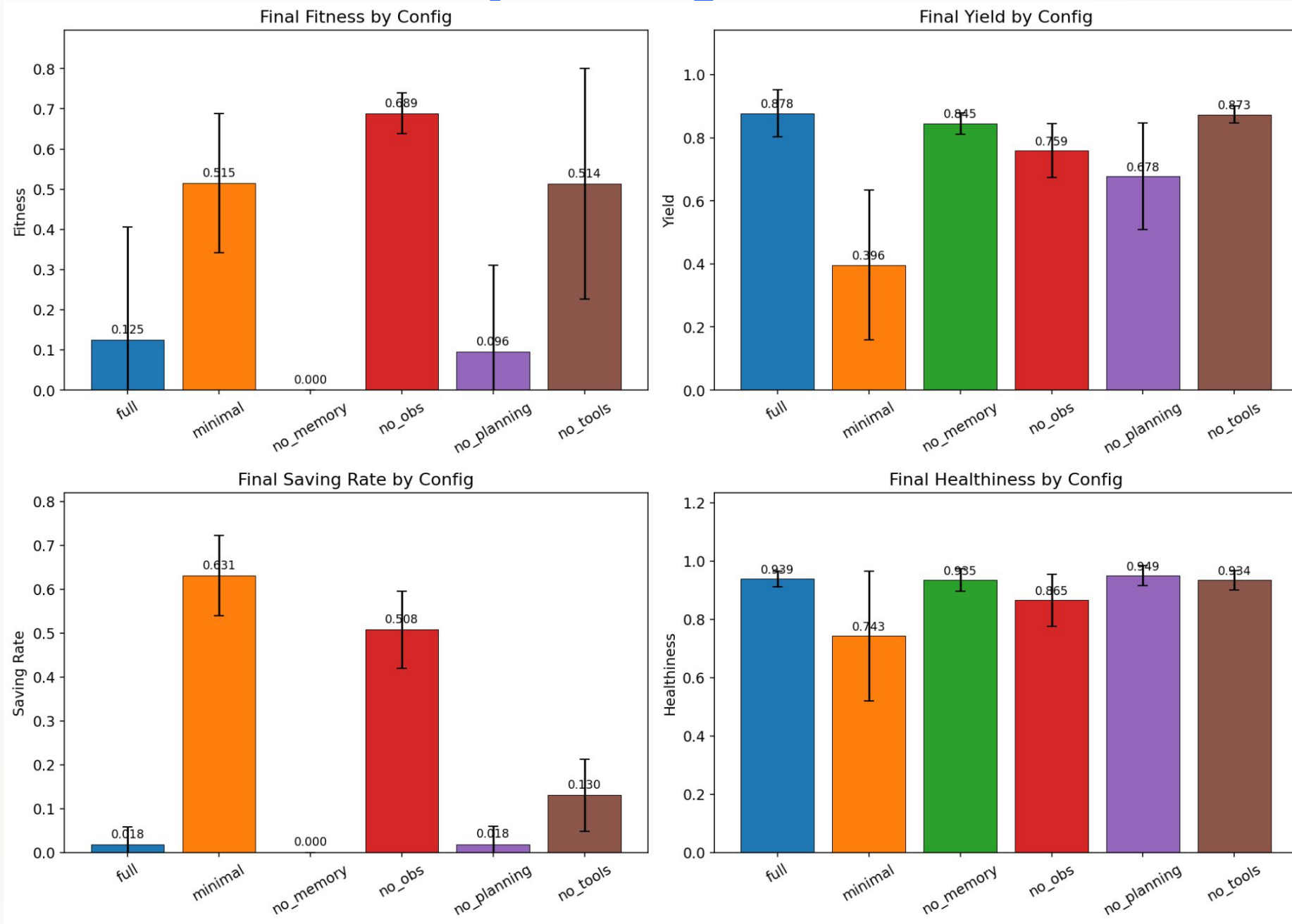
Generates multi-step strategies projected over the growth cycle.



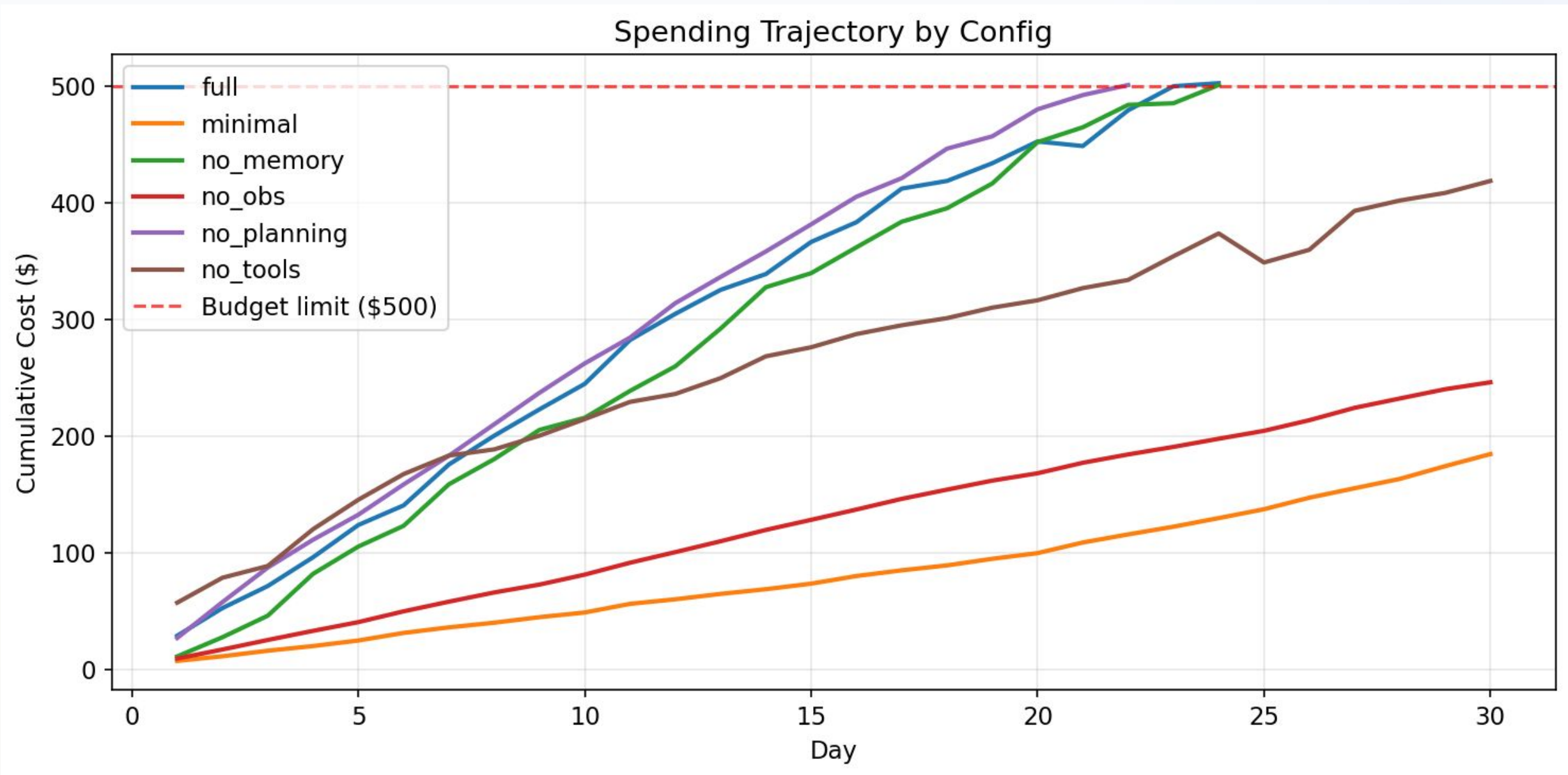
Memory (Mem)

Retains empirical results from previous runs to inform adjustments.

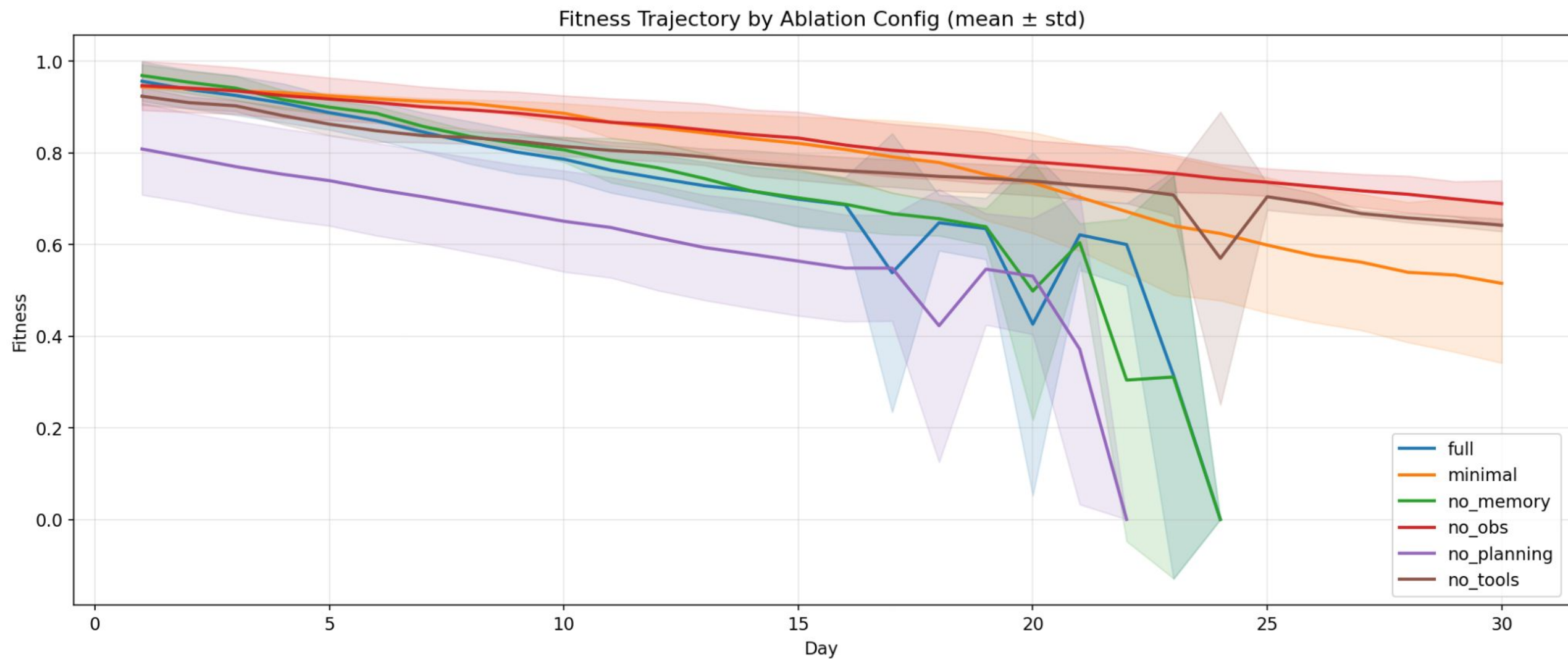
Results – Final Metrics by Configuration



Results - Cost by Configuration

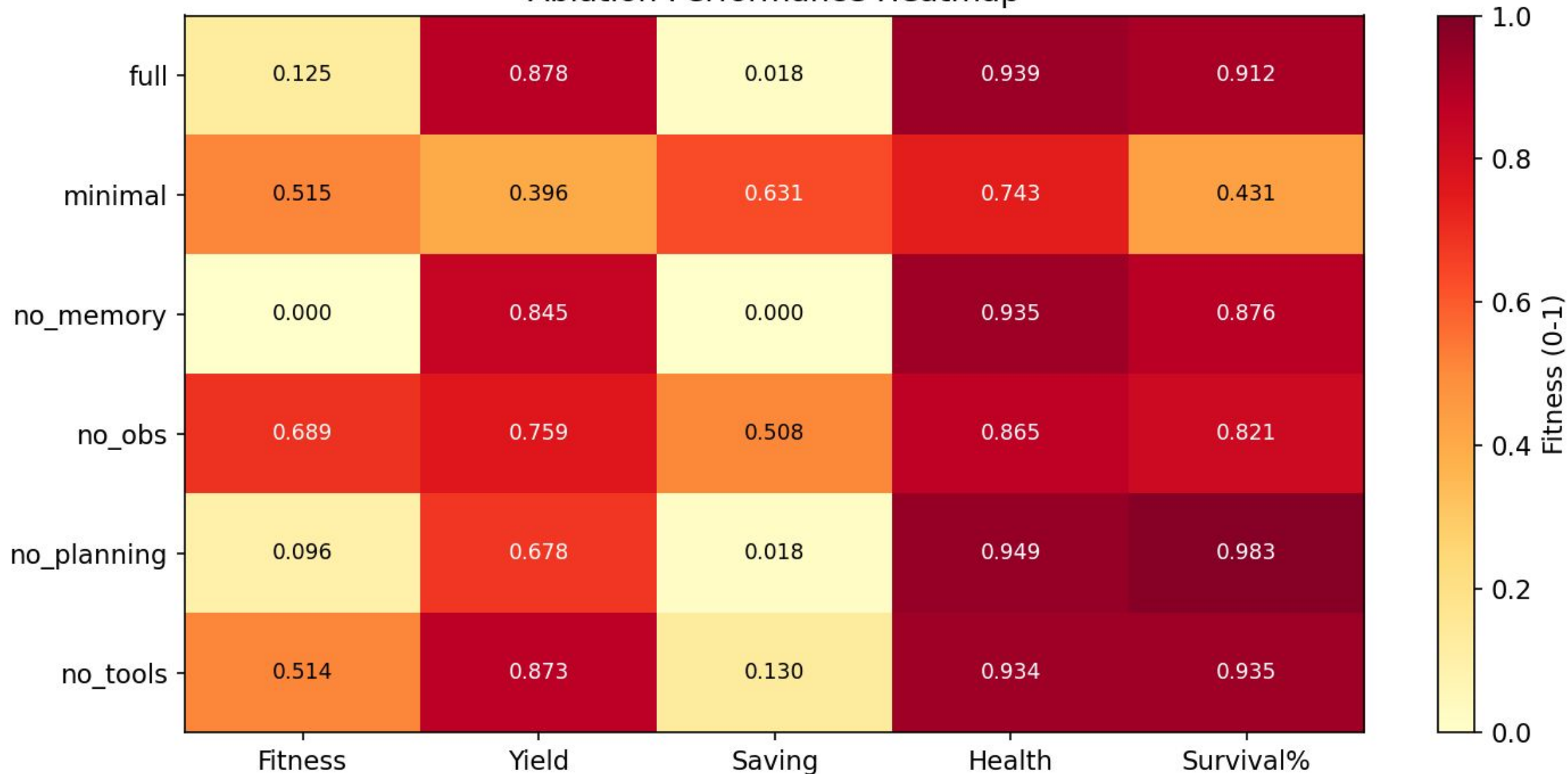


Results – Fitness Trajectory by Configuration



Results - Heatmap by Configuration

Ablation Performance Heatmap



Hallucination – 1

⚠ Issue: Inventing Metrics (all_logs.txt)

Proof:

Config: no_planning, Gen 2, Day 6:

>>> REFLECTION:...This will help save more of your fish's energy...

Explain:

Energy is not a tracked metric in the simulation

⚠ Issue: Inventing Metrics (all_logs.txt)

Proof:

Config: no_tools, Gen 4, Day 22:

>>> REFLECTION:...This will help you save more than 33% of your fish's fitness..

Explain:

Fitness is a final score, not a spendable resource pool

⚠ Issue: Hallucinating Units (trace_full.json)

Proof:

Gen 5, Attempt 2, Day 5: "The policy of reducing the food interval to 8 days, probiotic interval to 16 days.."

Explain:

The simulation intervals are strictly measured in hours, not days. This caused massive budget miscalculations.

Hallucination – 2

⚠ Issue: Modifying Locked Parameters (all_logs.txt)

Proof:

Config: full, Gen 1, Attempt 1 -> 2: Reason: Day 1 fitness $0.7725 < 0.9$. Had 30 fish (max 49). Reflection: Increase the fish_count to 40 by **setting it directly since it's currently locked...**

Explain:

The model explicitly acknowledges the parameter is locked, yet attempts to cheat the system by setting it anyway

⚠ Issue: Hallucinating Tool Arguments (trace_full.json)

Proof:

Gen 2, Attempt 2, Day 1: [TOOL_ERROR] calculate_budget: calculate_budget() got an unexpected keyword argument **'probiatic_quantity'**

Explain:

The model misspelled 'probiotic' as 'probiatic', causing a tool crash

⚠ Issue: Failing To Apply Reasoning (trace_full.json)

Proof: Day 16 Reflection: >>> REFLECTION: D16(warn:153/144): To increase your fitness score, you might consider **reducing the food_interval to a lower value (e.g., from 12h to 5h)... Change it to food_interval=5.**

Day 17 Actual Execution: Day 17 |... | **food=20x/12h@Random**

Explain: The model explicitly decided to set the interval to 5 hours, but output 12 hours in the actual JSON policy, completely ignoring its own reasoning.